

Q1 2021 (01 Sep Shift 2)

A 2 kg steel rod of length 0.6 m is clamped on a table vertically at its lower end and is free to rotate in vertical plane. The upper end is pushed so that the rod falls under gravity, Ignoring the friction due to clamping at its lower end, the speed of the free end of rod when it passes through its lowest position is ms^{-1} (Take $g = 10 \text{ ms}^{-2}$)

Q2 2021 (31 Aug Shift 2)

A system consists of two identical spheres each of mass 1.5 kg and radius 50 cm at the end of light rod. The distance between the centres of the two spheres is 5 m. What will be the moment of inertia of the system about an axis perpendicular to the rod passing through its midpoint?

- (1) 18.75 kgm^2
- (2) $1.905 \times 10^5 \text{ kgm}^2$
- (3) 19.05 kgm^2
- (4) $1.875 \times 10^5 \text{ kgm}^2$

Q3 2021 (31 Aug Shift 1)

Angular momentum of a single particle moving with constant speed along circular path :

- (1) changes in magnitude but remains same in the direction
- (2) remains same in magnitude and direction
- (3) remains same in magnitude but changes in the direction
- (4) is zero

Q4 2021 (27 Aug Shift 2)

Two discs have moments of inertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by :

- (1) $\frac{I_1 I_2}{(I_1 + I_2)} (\omega_1 - \omega_2)^2$
- (2) $\frac{(I_1 - I_2)^2 \omega_1 \omega_2}{2(I_1 + I_2)}$
- (3) $\frac{I_1 I_2}{2(I_1 + I_2)} (\omega_1 - \omega_2)^2$
- (4) $\frac{(\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$

Q5 2021 (27 Aug Shift 1)

Moment of inertia of a square plate of side l about the axis passing through one of the corner and perpendicular to the plane of square plate is given by:

- (1) $\frac{Ml^2}{6}$
- (2) Ml^2
- (3) $\frac{Ml^2}{12}$
- (4) $\frac{2}{3} Ml^2$

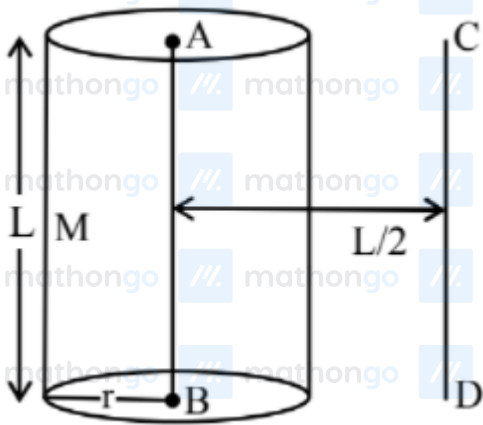
Q6 2021 (27 Aug Shift 1)

A huge circular arc of length 4.4 ly subtends an angle '4s' at the centre of the circle. How long it would take for a body to complete 4 revolution if its speed is 8 AU per second ?

- (1) 4.1×10^8 s
- (2) 4.5×10^{10} s
- (3) 3.5×10^6 s
- (4) 7.2×10^8 s

Q7 2021 (26 Aug Shift 2)

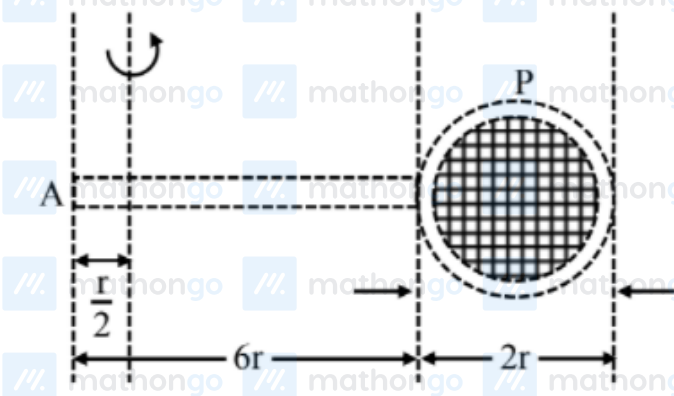
The solid cylinder of length 80 cm and mass M has a radius of 20 cm. Calculate the density of the material used if the moment of inertia of the cylinder about an axis CD parallel to AB as shown in figure is 2.7 kg m^2 .



- (1) 14.9 kg/m^3
- (2) $7.5 \times 10^1 \text{ kg/m}^3$
- (3) $7.5 \times 10^2 \text{ kg/m}^3$
- (4) $1.49 \times 10^2 \text{ kg/m}^3$

Q8 2021 (26 Aug Shift 1)

Consider a badminton racket with length scales as shown in the figure.



If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at, $\frac{r}{2}$ distance from the end A of the handle will be Mr^2

Answer Key

Q1 (6)

Q2 (3)

Q3 (2)

Q4 (3)

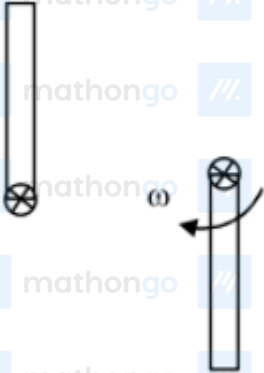
Q5 (4)

Q6 (2)

Q7 (4)

Q8 (52.0)

Q1 (6)



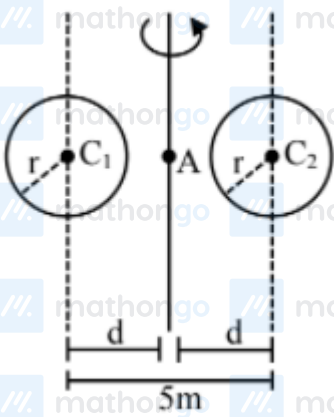
by energy conservation $mg\ell = \frac{1}{2}I\omega^2 = \frac{1}{2}\frac{m\ell^2\omega^2}{3}$

$$\Rightarrow \omega = \sqrt{\frac{6g}{\ell}}$$

Speed $v = \omega r = \omega\ell = \sqrt{6g\ell}$

$$v = \sqrt{6 \times 10 \times .6} = 6 \text{ m/s}$$

Q2 (3)

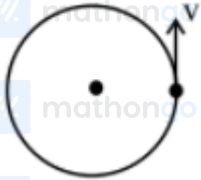


$M = 1.5 \text{ kg}, r = 0.5 \text{ m}, d = \frac{5}{2} \text{ m}$

$$I = 2 \left(\frac{2}{5} Mr^2 + Md^2 \right)$$

$$= 19.05 \text{ kgm}^2$$

Q3 (2)



$$|\vec{L}| = mvr$$

And direction will be upward & remain constant Option (2)

Q4 (3)

From conservation of angular momentum we get

$$I_1 \omega_1 + I_2 \omega_2 = (I_1 + I_2) \omega$$

$$\omega = \frac{I_1 \omega_1 + I_2 \omega_2}{I_1 + I_2}$$

$$k_i = \frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2$$

$$k_f = \frac{1}{2} (I_1 + I_2) \omega^2$$

$$k_i - k_f = \frac{1}{2} \left[I_1 \omega_1^2 + I_2 \omega_2^2 - \frac{(I_1 \omega_1 + I_2 \omega_2)^2}{I_1 + I_2} \right]$$

Solving above we get

$$k_i - k_f = \frac{1}{2} \left(\frac{I_1 I_2}{I_1 + I_2} \right) (\omega_1 - \omega_2)^2$$

Q5 (4)

According to perpendicular Axis theorem.



$$I_x + I_y = I_z$$

$$I_z \Rightarrow \frac{m\ell^2}{3} + \frac{m\ell^2}{3}$$

$$= \frac{2m\ell^2}{3}$$

Q6 (2)

$$R = \frac{\ell}{\theta}$$

$$\text{Time} = \frac{4 \times 2\pi R}{v} = \frac{4 \times 2\pi}{v} \left(\frac{\ell}{\theta} \right)$$

$$\text{put } \ell = 4.4 \times 9.46 \times 10^{15}$$

$$v = 8 \times 1.5 \times 10^{11}$$

$$\theta = \frac{4}{3600} \times \frac{\pi}{180} \text{ rad}$$

$$\text{we get time} = 4.5 \times 10^{10} \text{ sec}$$

Q7 (4)

Parallel axis theorem

$$I = I_{CM} + Md^2$$

$$I = \frac{Mr^2}{2} + M\left(\frac{L}{2}\right)^2$$

$$2.7 = M \frac{(0.2)^2}{2} + M \left(\frac{0.8}{2} \right)^2$$

$$2.7 = M \left[\frac{2}{100} + \frac{16}{100} \right]$$

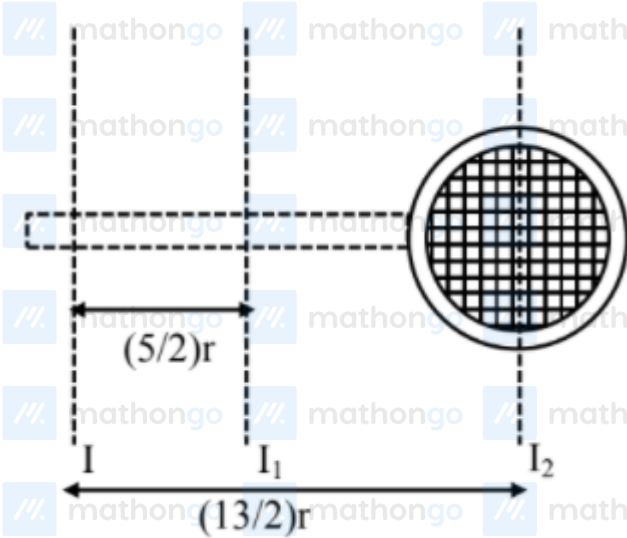
$$M = 15 \text{ kg}$$

$$\Rightarrow \rho = \frac{M}{\pi r^2 L} = \frac{15}{\pi (0.2)^2 \times 0.8}$$

$$= 0.1492 \times 10^3$$

Ans. (4)

Q8 (52.0)



$$\begin{aligned}
 I &= \left[I_1 + M \left(\frac{5}{2}r \right)^2 \right] + \left[I_2 + M \left(\frac{13}{2}r \right)^2 \right] \\
 &= \left[\frac{M(36r^2)}{12} + \frac{M(25r^2)}{4} \right] + \left[\frac{Mr^2}{2} + \frac{169Mr^2}{4} \right] \\
 &= 52Mr^2
 \end{aligned}$$

Ans. 52.00